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Max Montes Soza

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Employment

Software Engineer	Microsoft Corporation	Fall 2025 – Present
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- Engineered and operated Partner Center transaction services processing **1.1 billion requests, 1M purchases, and 3.8M subscription updates per month** with **over 99.9% availability**, supporting Microsoft's **\$35B/year partner network**.
- Owned frontend UX for a Partner Center promotions launch spanning **20 teams**, resolving cross-team integration and launch-blocking bugs under tight deadlines. React, TypeScript, C#.
- Performed on-call duties for transaction services, owning incident triage, mitigation, **root-cause analysis**, and **monitoring**; built an AI-assisted tool that reduced weekly production-health reporting from hours to **~5 minutes**.
- Built AIAssist Copilot features spanning **RAG**, prompt engineering, API integration, UI, and monitoring for natural-language analytics; established **evaluation, feedback loops, and LLM guardrails** to improve AI output reliability. C#, Azure.
- Developed timer-triggered Azure Functions for report generation and promotion eligibility, including subscription mutations and webhooks; provisioned production storage, secrets, **observability and alerting infrastructure**.

SWE Intern	Microsoft Corporation	Summer 2024
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- Engineered a Copilot plugin automating **~70 daily support tickets (~\$75 per manual review)**, using **RAG and vector embeddings** for context-aware retrieval. C#, Cosmos DB.
- Built an **orchestrated AI agent system** with Semantic Kernel, integrating APIs to coordinate agentic workflows and LLM-powered tool use.

Education

Master's	Georgia Institute of Technology	2026 – Present
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- M.S. in Computer Science, **Machine Learning** specialization. **GPA: 4.0**. Coursework: Machine Learning, Deep Learning, Computer Vision.

Undergraduate	Boston College	Fall 2021 – May 2025
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- B.A. in Computer Science, B.A. in Philosophy. **GPA: 3.7, Cum Laude**. Coursework: Algorithms, Operating Systems, Computer Networks.

Projects

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- Art Explorer** — Built **multimodal semantic search** across **5,500 Met paintings** using SigLIP text-image embeddings, PostgreSQL/pgvector HNSW indexes, and hybrid retrieval; added resumable ingestion and human relevance evaluation. TypeScript, Next.js, Python.
 - Benchmarking Supervised Learning Algorithms** — Evaluated DT, kNN, SVM, and neural networks across **45K+ samples**; implemented custom **PyTorch** MLPs and optimized hyperparameters with Optuna. Python, Scikit-learn.

Additional Experience and Awards

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- IEEE Publication:** Researched and published a bibliographic format convention that improved read-write performance by **26%**.

Languages and Technologies

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- Languages:** Python, C/C++, Java, JavaScript, C#, TypeScript, SQL, Rust.
 - ML / AI:** LLMs, Generative AI, RAG, Vector Embeddings, PyTorch, Scikit-learn, Semantic Kernel, Prompt Engineering.
 - Cloud / Engineering:** Azure, Azure Functions, FastAPI, React, REST APIs, Observability, Linux, Git.